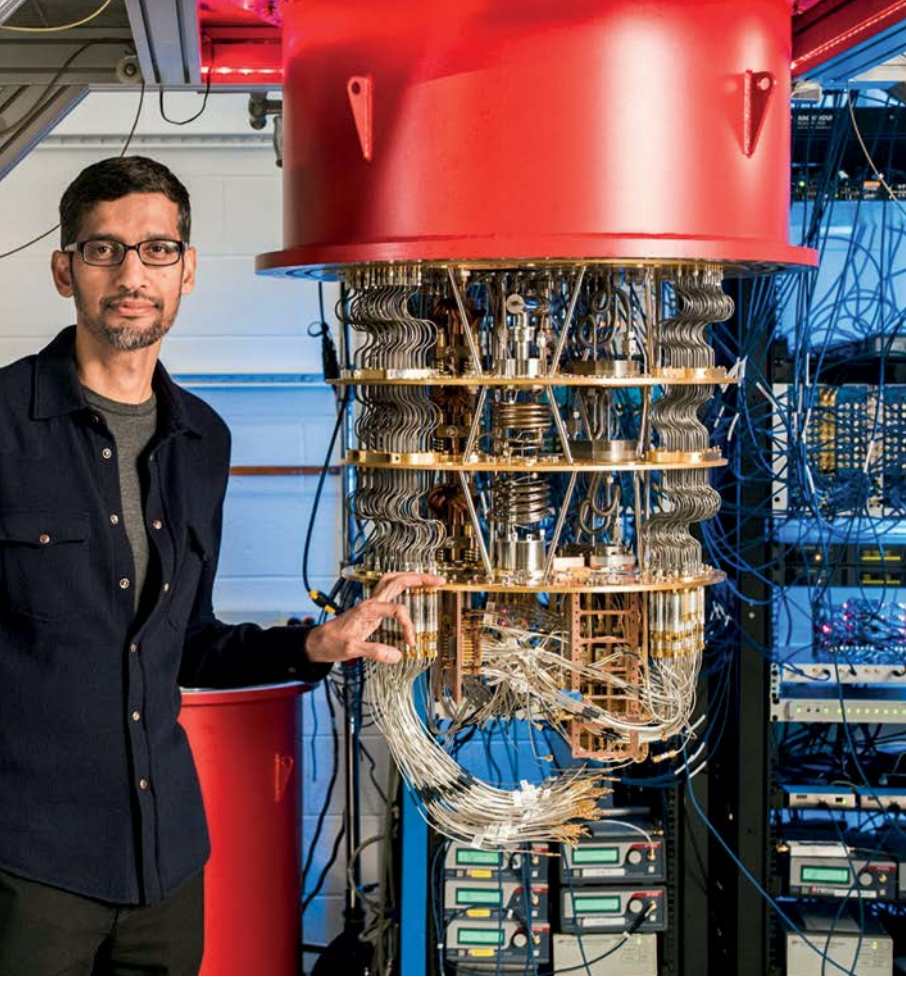


REUTERS



"Running full simulation models of an airplane with high fidelity requires high computing power. What we are doing is plugging quantum algorithms into these simulations to be able to better utilise the hardware that you have"

– ABHISHEK CHOPRA, *Founder, BosonQ Psi*

critical. Next on the horizon are quantum sensors to measure electric, magnetic or gravitational fields, time or space. These sensors could make MRI machines used in medical diagnosis much more sensitive and smaller, perhaps even portable, says Prof. Apoorva D. Patel of the Center for Excellence in Quantum Technology at the Indian Institute of Science, Bengaluru.

WHAT INDIA NEEDS TO MASTER

When it comes to quantum hardware, India is at least five years behind global leaders. IBM's latest quantum processor, Osprey, has a qubit count of 433, with plans to scale up to 4,000+ by 2025. In April, India approved a National Quantum Mission targeting 50-100 qubit computers in five years and up to 1,000 qubits in eight years. Other focus areas are quantum communications, quantum sensors and metrology, and quantum materials and devices.

"Essentially, India will have to build expertise bottom-up, as it did with nuclear and space tech," says Prof. Patel. This means developing simulators and tools currently available only with a few companies globally. The mission is long-term, but the leap will be a quantum one. ■

THE CHANGE-MAKERS

BosonQ Psi

- Founded in 2020, the Bengaluru-headquartered start-up works on quantum algorithms for simulations used by aerospace and automotive companies
- Quantum algorithms can solve certain optimisation problems faster than classical algorithms which use up a lot of computing power to perform the same task
- It is currently working on various proofs-of-concept in design optimisation, such as weight reduction for a car chassis or an electric vehicle battery

QULABZ

- Founded by Nixon Patel, who is also its CEO and an adjunct professor at IIT Hyderabad, Qulabz is currently working on quantum communication technologies
- These include quantum random number generators, touted to be more secure than the OTPs today, and a quantum network simulator, which can design a secure communication link end to end
- The third product is a portfolio-optimising tool for financial investments using quantum techniques and Artificial Intelligence